Tejdeep Kellampalli

Guntur, Andhra Pradesh | +91 93983 07247 | kellampallitejdeep@gmail.com
linkedin.com/in/tejdeep-kellampalli-6510a8355 | github.com/Tejdeep-67 | tejdeep-portfolio.netlify.app

SUMMARY

Final-year B.Tech CSE (Data Science) student (2027) who builds real-time data products end to end: live API data pipelines, **SQL** analysis on 1M+ rows, and **Python ML** models tested against baselines on unseen data and deployed as working web apps. Seeking Data/Business Analyst, QA or IT internships and entry-level roles (Hyderabad, Bangalore, Chennai, remote).

TECHNICAL SKILLS

- **Languages:** Python, SQL, C, JavaScript / TypeScript (Basics)
- **Data & ML:** pandas, NumPy, scikit-learn, LightGBM, Matplotlib, Seaborn, EDA, Forecasting, NLP (TF-IDF), ONNX
- **BI & Tools:** Power BI, Excel, Jupyter Notebook, Git, GitHub, pytest
- **Web & Databases:** HTML, CSS, React.js, Flask, REST APIs, Leaflet, Netlify, SQLite, MongoDB

PROJECTS

Safe Roads | *React, TypeScript, YOLOv8, ONNX Runtime Web, Leaflet, Flask* Live | GitHub

- Built real-time, in-browser pothole detection: a YOLOv8n-seg model exported to **ONNX (14 MB)** and run in a Web Worker (WebGPU/WASM) on webcam, video and photos; an IoU tracker counts each pothole once.
- Benchmarked the model (fine-tuned on a public dataset) on **100 unseen Roboflow test images** (269 potholes): **mAP@50 0.34, precision 0.68** at 640 px; documented low recall on small potholes.
- Added a shared hazard map (Netlify Functions + Blobs, validation, rate limiting) and route alerts for hazards **within 30 m** of an OSRM route with a voice/vibration drive mode; Flask API covered by **20 pytest tests**.

AirPulse | *React, Python, scikit-learn, LightGBM, Open-Meteo API* Live | GitHub

- Built a live air quality and weather monitor for **37 Indian cities** that computes the CPCB National AQI in code from Open-Meteo (CAMS) data, refreshed every 15 minutes.
- Developed a 24-hour PM2.5 forecast (ridge + LightGBM blend) from 22 months of hourly data; on unseen test months (Sep–Oct 2026), **MAE 7.72 vs 11.67 µg/m³** for persistence (**34% lower**), better in all 37 cities.
- Runs the forecast in the browser, with AQI alerts as browser notifications and an insights page (wind vs PM2.5).

NiftyLens | *React, Netlify Functions, Python, scikit-learn* Live | GitHub

- Built a live NIFTY 50 dashboard: a cached serverless proxy for quotes, candlestick charts with SMA/RSI, watchlist, portfolio P&L and price alerts.
- Trained a TF-IDF + logistic regression headline sentiment model: **macro-F1 0.821** on held-out Indian financial headlines (SEntFiN) vs 0.716 for zero-shot FinBERT.
- Tested whether news sentiment predicts next-day returns on **2,310 stock-days** (**17,172 headlines**): no signal (r ≈ 0.00), reported as an honest null result.

Retail Sales Analytics | *SQL, SQLite, Python, ECharts* Live | GitHub

- Cleaned **1.07M** raw retail transactions (duplicates, cancellations, non-product codes) into a 5-table SQLite database.
- Used window functions (LAG, NTILE) for MoM growth, RFM segments and cohort retention; found the top 20% of customers drive **77% of revenue** and flagged **829 at-risk customers worth £1.5M**.
- Built a Power BI-style dashboard with KPI tiles, slicers and cross-filtering.

EDUCATION

R.V.R. & J.C. College of Engineering, Guntur — Sep 2023 – Jun 2027
B.Tech in Computer Science and Engineering (Data Science) — CGPA: 7.94/10
Sri Chaitanya Junior College | Intermediate (MPC): 79.5% — Jun 2021 – Apr 2022
Sri Chaitanya School | SSC: 100% — Jun 2020 – Apr 2021

CERTIFICATIONS

- Certified Data Science Professional – Oracle
- Certified Gen AI Professional – Oracle
- AICTE Google AI-ML Virtual Internship – EduSkills
- AICTE Microsoft Foundations of AI Virtual Internship – Edunet Foundation